A high-speed digital equaliser for satellite applications

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This paper reviews the design of a very high speed digital filter for removal of unwanted amplitude and group-delay distortion from signals that are transmitted over a satellite link.

This equalisation process, implemented with a high-speed digital filter, is easy to configure and stable. Desired channel characteristics are entered in a table and the equaliser can be reconfigured by the push of a button.

1. Introduction

An earth station is the part of a satellite system that can transmit or receive signals from a satellite in space, as shown in Fig 1. The satellite earth stations operated by BT can be large and complex, with cross-site cables and redundancy switching. This, together with the frequency up-conversion processes, antenna, RF equipment and the satellite link, introduces amplitude and group delay distortions. The BT earth stations currently use analogue equalisers to reduce these distortions.

However, a number of problems exist with the current equaliser equipment, primarily due to the particular analogue implementation. These problems result in making the equalisers difficult and time consuming to set up. The analogue components are manually adjusted, which can take a number of days and repeating previous results can be difficult. The analogue components are prone to drift and ageing.

An equaliser can be implemented using a digital finite impulse response (FIR) filter. Digital equalisation is easy to configure and stable. Once the desired channel characteristics are entered in a software table, the equaliser

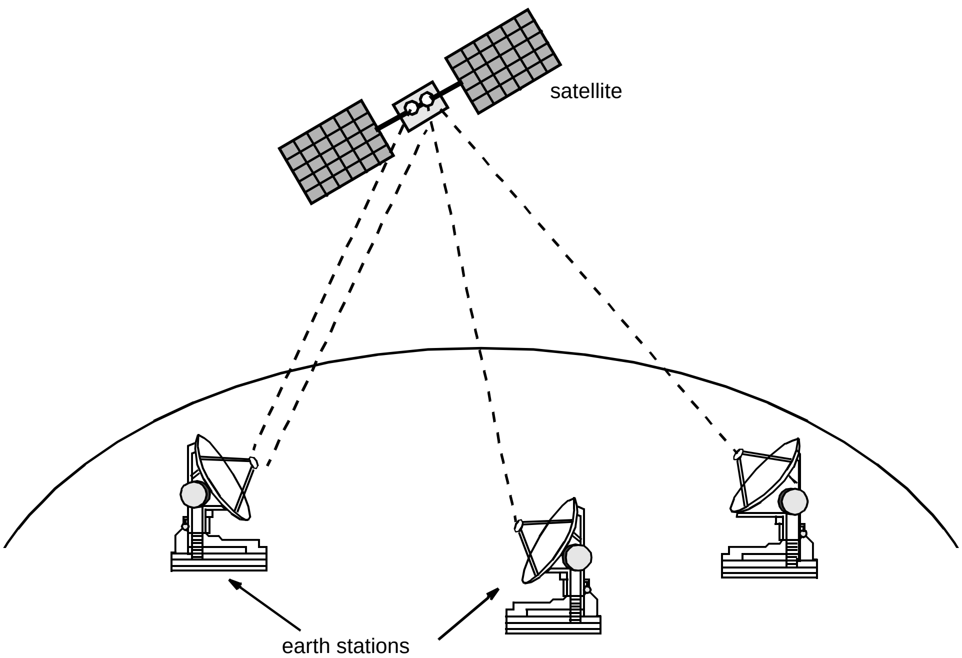

Fig 1 Earth stations transmitting/receiving signals from a satellite.

can be reconfigured by the push of a button. Results can be accurately repeated and the equaliser is not prone to drift and ageing. The digital equaliser specification is:

- input spectrum of 140 MHz ± 40 MHz,
- equalisation to 0.5 dB for centre 48 MHz, and 1 dB for the rest to 72 MHz amplitude response, and 3 ns for centre 60 MHz and 8 ns for the rest to 72 MHz group delay response (see Fig 2),
- negligible degradation in expected bit error rate performance.

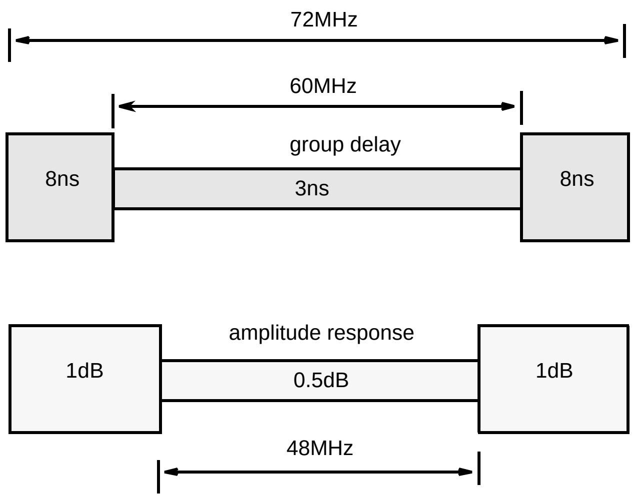

Fig 2 Equalisation mask.

A digital equaliser will be invaluable for the installation of cross-site cables and equipment for high bandwidth signals such as the Intelsat IDR and TDMA services. The digital equaliser will be used for a cable restoration service using a 140 Mbit/s system. Transatlantic cables can become faulty, e.g. damaged by trawlers; the repair is performed by cable ships and is time consuming. The service is restored via satellite within 2 hours and this service can be provided via satellite for several weeks, saving millions of pounds a day in lost revenue.

2. Finite impulse response filter

A FIR filter has a digital input and the output is formed by taking a weighted sum over its value at equally spaced times as shown in Fig 3. The filter has the interesting property of being symmetrical in time, i.e. averaging the past and future to arrive at the present output.

The digital input, Di, is weighted at equally spaced times, by the coefficient values C1...C7. The number of equally spaced times, generated using delay elements, are known as taps.

3. System requirements

A study determined that the necessary digital signal processing technology was available. Software simulation using a block-oriented simulation system (BOSS) was used in the feasibility study to model the specified worst case intermediate frequency (IF) channel distortions and the equalisation provided by the proposed implementation for the digital equaliser. The simulation results demonstrated that a 90-tap FIR filter would provide sufficient equalisation to achieve the required level for the 80-MHz bandwidth channels. Frequency down-conversion and up-conversion lower the required sample rate and hence the complexity of the digital circuitry. A digital low-speed model was fabricated using transistor-transistor logic (TTL) to prove the concept.

Software was written for entering channel characteristics, computing the required equaliser response and programming the equaliser. This may be later modified to automatically capture channel characteristics from a network analyser and provide automatic equalisation.

4. System design

4.1 *User interface*

The operator first measures the amplitude and group delay characteristics of the path to be equalised using standard techniques. The data is then entered into a table using a personal computer (PC). If a simple flat

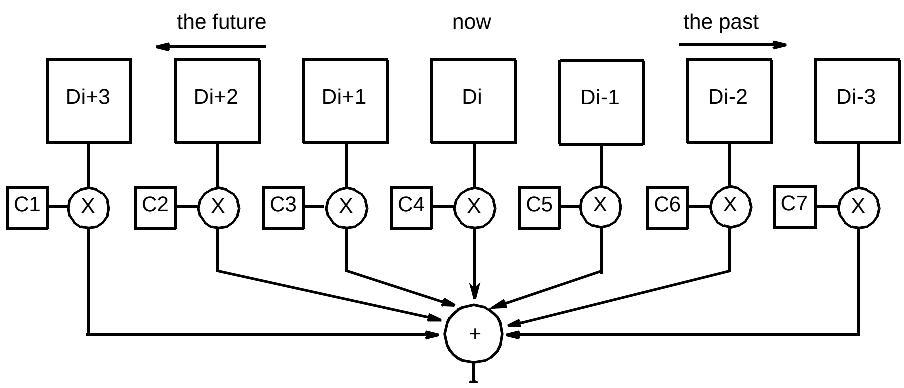

Fig 3 Digital finite impulse response filter with seven taps.

response is required from the equaliser, the PC is commanded to adjust the digital equaliser accordingly. There may be occasions where pre-distortion is required in order to equalise the characteristics of earth station or satellite path effects following the equaliser, and this can also be readily programmed.

The current user interface software runs on a PC under the Microsoft Windows ™ operating environment. This permits the use of the standard interface and controls with which people are familiar. The software has been written in C++ using an object oriented approach. Data entered and the results of calculations are stored in files on the PC's hard disk. Automated testing of the hardware and in particular the filter modules, has been incorporated into the application.

4.2 Algorithm development for the digital equaliser

The algorithms for coefficient calculation operate by combining the measured channel characteristic with the desired frequency response, and converting these into the desired impulse response for the equaliser. The digital equaliser acts as a finite impulse response filter, with the duration of the impulse response dictated by the sample rate and the number of coefficients the filter is capable of supporting.

The desired impulse response is windowed using the duration of the filter as a constraint, the shape of the window being varied to minimise the mean square error of the combined frequency response in the frequency range specified, as shown in Fig 4. Quantisation of the coefficients, to an accuracy that can be supported by the equaliser hardware, generates distortion of the frequency response. Adjustment of the window shape and position allows some compensation for this quantisation distortion.

Modification of the window shape affects different portions of the frequency spectrum. Making the window shape more rectangular increases the magnitude of the ripple in the amplitude and group delay responses within the frequency domain. However, making the window too smooth can cause significant loss of energy at the ends of the impulse response, and if the group variation across the frequency band is large then this can cause significant reduction in the amplitude response at the edges of the frequency band. Adjusting the window shape adjusts the trade-off between these two effects — the optimum value will be different for each frequency response.

Future versions of the software will automatically adjust the individual coefficients to attempt to reduce the error signal in the response to fit within an arbitrary mask as specified by the user.

4.3 Processing platform

The digital equaliser consists of a signal processing engine operating at 240 million samples per second. Due to the very high sampling rate and relatively large number of coefficients (taps) required, the FIR filter function is implemented using a novel parallel processing architecture.

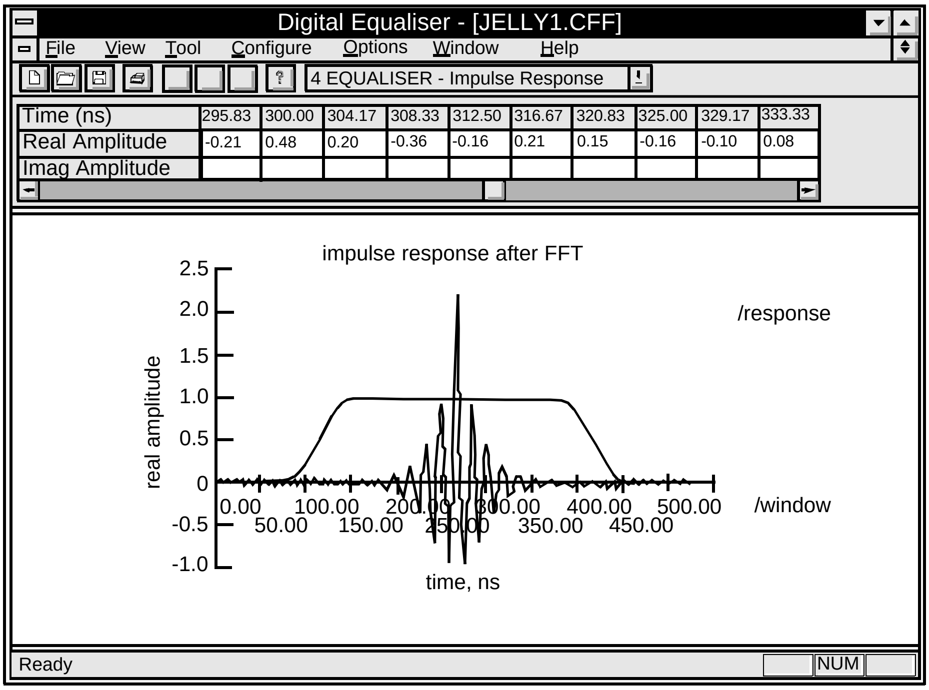

Fig 4 The windowed impulse response.

The analogue input is converted to an 8-bit wide word at 240 million samples per second. This digital data is demultiplexed into multiple parallel data streams at a lower bit rate, for ease of filtering. Following the filter process, the data streams are multiplexed to form a 12-bit word at 240 million samples per second which is processed by a high-speed digital-to-analogue converter (see Fig 5).

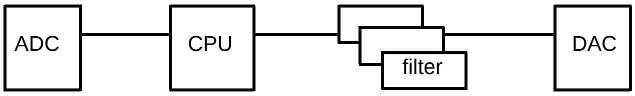

Fig 5 The filter process.

The filter process is of a modular construction, using between 12 and 60 VLSI matrix multiplier integrated circuits (ICs), representing filter lengths between 18 and 90 taps. Each multiplier IC contains nine 10 × 12-bit digital multiplier and three 16-bit adders. The multiplier ICs are clocked synchronously at 40 MHz, and in the case of a 90-tap filter, perform 21.6 billion arithmetic multiplications per second.

The equaliser is controlled by a CPU card, which is based on a Motorola M68000 microprocessor. The CPU card passes the data from the analogue-to-digital converter to the filter modules. A second CPU card can be placed following the filter modules, for testing purposes.

The main function of the CPU card is to load the selected coefficients into the filter modules. This is controlled by the operator using the front-panel interface or by a PC connected to the serial interface. The serial interface allows equaliser characteristics to be transferred to the CPU card and stored in battery-backed RAM.

5. Implementation issues

There were many implementation issues that had to be addressed in the design of this very high-speed digital equaliser. In brief, some of these included:

- the design layout for the analogue reference circuitry and ground planes associated with the analogue-to-digital converter were critical — incorrect layout would have resulted in 'noise' pick-up on the analogue input,

- the PCB tracks carrying data at rates up to 240 million bits per second required treatment as transmission lines, with due consideration of characteristic impedance and termination effects — the number of diversions and changes in track direction were kept to a minimum by hand-routeing the PCBs,

- the tracks carrying high-speed data buses needed to be very similar in length to avoid timing errors due to data skewing,

- as many of the components operate near their maximum timing specifications and on differing clock edges, the generation of clocks with symmetrical mark space ratios was critical,

- layout of the boards in general was very critical to eliminate clock breakthrough,

- the boards were multilayer (up to ten layers), including solid power and ground layers to minimise the length of earth return loops — analogue power supplies were independently generated and distributed,

- careful selection of inter-board connectors was also essential to reduce possible crosstalk effects between signals — this was assisted by adequate signal-to-earth spacing on the connector's signal allocations.

6. Tests

Performance testing includes group delay and amplitude equalisation capability, spurious generation, noise floor levels, third-order intercept levels, and bit error rates under various test conditions with 45 Mbit/s, 120 Mbit/s, 140 Mbit/s and 155 Mbit/s carriers.

The performance of the equaliser was compared to simulated results for amplitude response and group delay. The results recorded matched the simulated results. With the equaliser configured for 90 taps of filtering, an arbitrary amplitude response was specified in the user interface software, as shown in Fig 6, and the resulting output from the equaliser was captured on a network analyser, as shown in Fig 7.

7. Conclusions

An equaliser can be implemented using a complex digital filter with a bandwidth of 80 MHz. A digital equaliser has been developed which provides significant advantages over the existing analogue equalisers currently in use at BT's earth stations.

- the digital equaliser maintains the programmed response very accurately with negligible susceptibility to component values changing due to ageing,

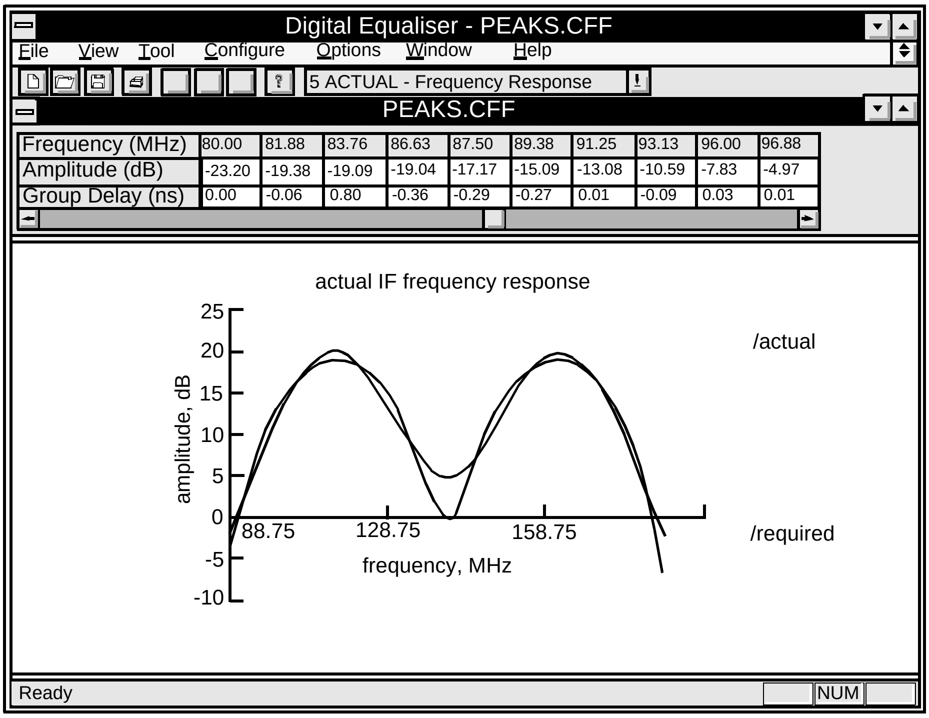

Fig 6 Amplitude response.

- programming of the equaliser for any required characteristic is fast and very straightforward, significantly reducing set-up times,

- various equalisation characteristics can be stored and selected locally via front panel switches,

- pre-distortion can be easily implemented, in order to equalise the characteristics of earth station or satellite path effects following the equaliser,

- the digital equaliser lends itself to future developments which may include an adaptive response to changing path characteristics.

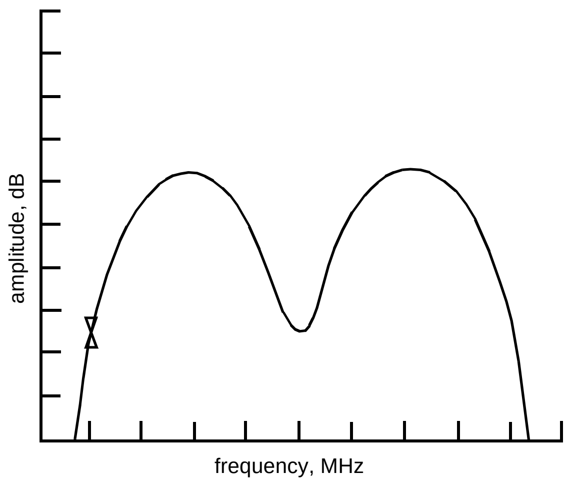

Fig 7 Equaliser output.

The digital equaliser has successfully performed a number of tests with a 140 Mbit/s modem in an IF loop and over a satellite link carrying live test traffic.

The equaliser performs digital arithmetic instruction like a PC; however, it has a fixed instruction set. A fast PC of 90 MHz performs 15 million instructions per second. The digital equaliser performs 21 billion instructions per second. The equaliser has a novel parallel architecture which allows it to perform functions at this speed.

The product prototype has 90 taps in a modular approach which allows for more or less taps in each unit. The hardware uses a novel approach of no backplane — instead complex connectors carry 140 signal-plus-ground connections from board to board. The boards are hand-routed, because of the speed of signals, with up to ten layers. The integrated circuits used are working to the limit of their specification.

Helen Rhodes joined BT Laboratories in 1980. She has worked for the majority of this time in the field of digital design of hardware. Work has included involvement in design and construction of equipment for bandwidth efficient TV systems, and participation in transatlantic satellite trials. In 1991 Helen was a runner-up in the young woman engineer of the year IEE competition.

Since 1993 she has had the major responsibility for hardware design and testing of a 240 mega sample per second 90-tap digital equaliser.